

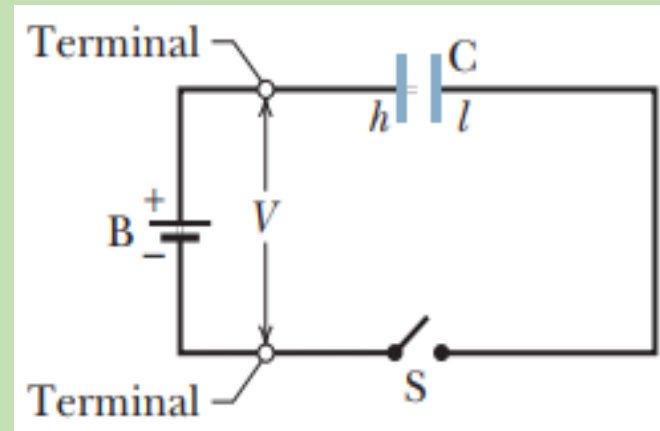
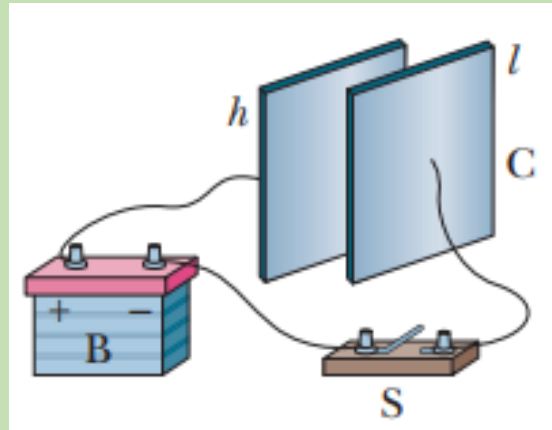
Lecture 11

Chapter 25
Capacitance

□ Capacitance

Capacitance is the ability of a device to store energy in the form of electrical charge

This device is called Capacitor.



When a capacitor is charged, its plates have charges of equal magnitudes but opposite signs: $+q$ and $-q$

The charge q stored in the capacitor is proportional to the potential difference V between the two conductors.

$$q \propto V \\ \Rightarrow q = CV$$

The proportionality constant C is called the capacitance of the capacitor.

Unit: Farad, F

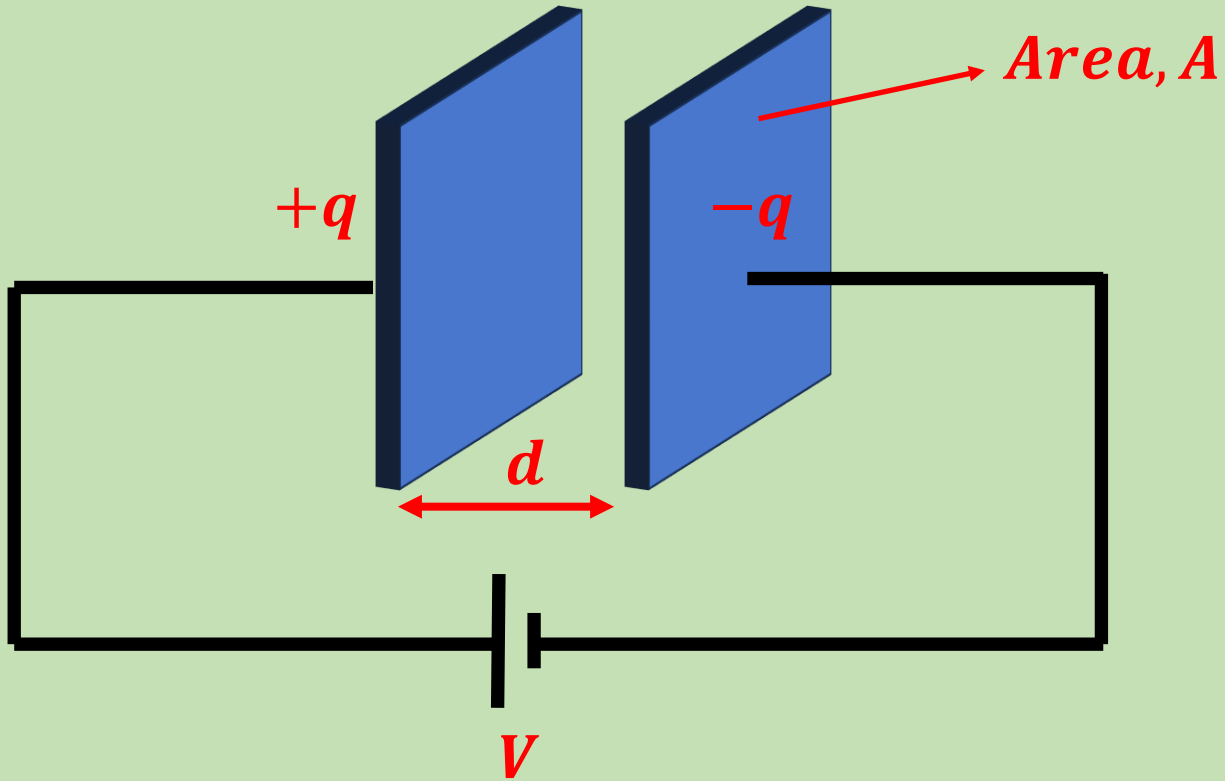
➤ *Charging of capacitor*

When the switch is closed, electrons are driven through the wires by an electric field that the battery sets up in the wires. The field drives electrons from capacitor plate ***h*** to the positive terminal of the battery; thus, plate ***h***, losing electrons, becomes positively charged. The field drives just as many electrons from the negative terminal of the battery to capacitor plate ***l***; thus, plate ***l***, gaining electrons, becomes negatively charged just as much as plate ***h***, losing electrons, becomes positively charged.

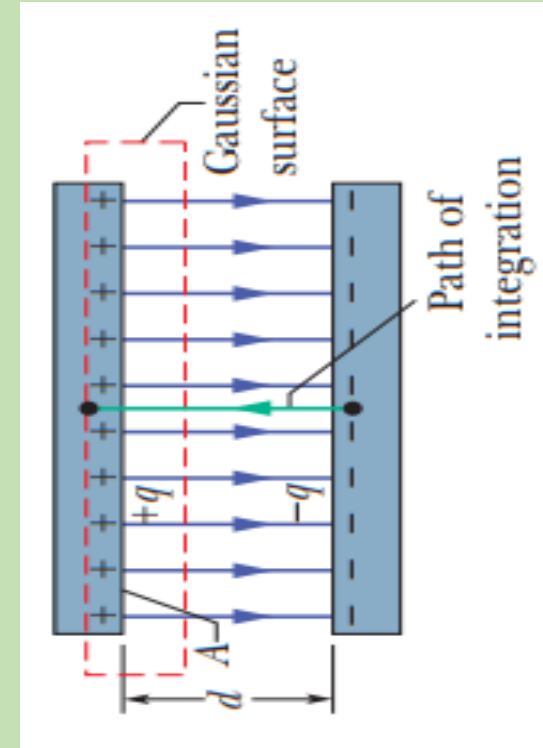
Initially, when the plates are uncharged, the potential difference between them is zero. As the plates become oppositely charged, that potential difference increases until it equals the potential difference ***V*** between the terminals of the battery. Then plate ***h*** and the positive terminal of the battery are at the same potential, and there is no longer an electric field in the wire between them. Similarly, plate ***l*** and the negative terminal reach the same potential, and there is then no electric field in the wire between them. Thus, with the field zero, there is no further drive of electrons. The capacitor is then said to be fully charged, with a potential difference ***V*** and charge ***q*** that are related by the equation, **$q = CV$** .

□ Calculating the Capacitance

❖ Parallel plate Capacitor

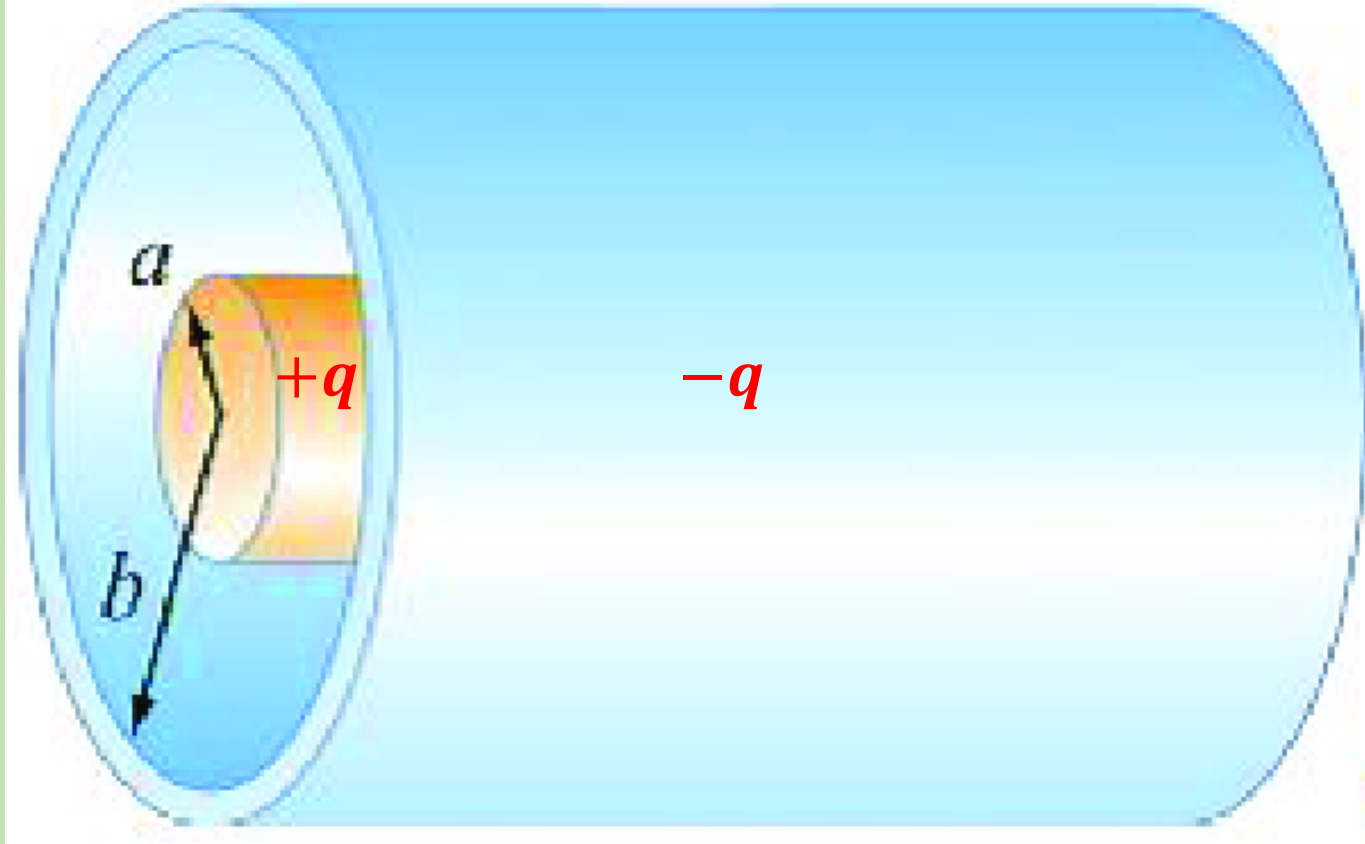


$$\text{Capacitance, } C = \frac{\epsilon_0 A}{d}$$



❖ Cylindrical Capacitor

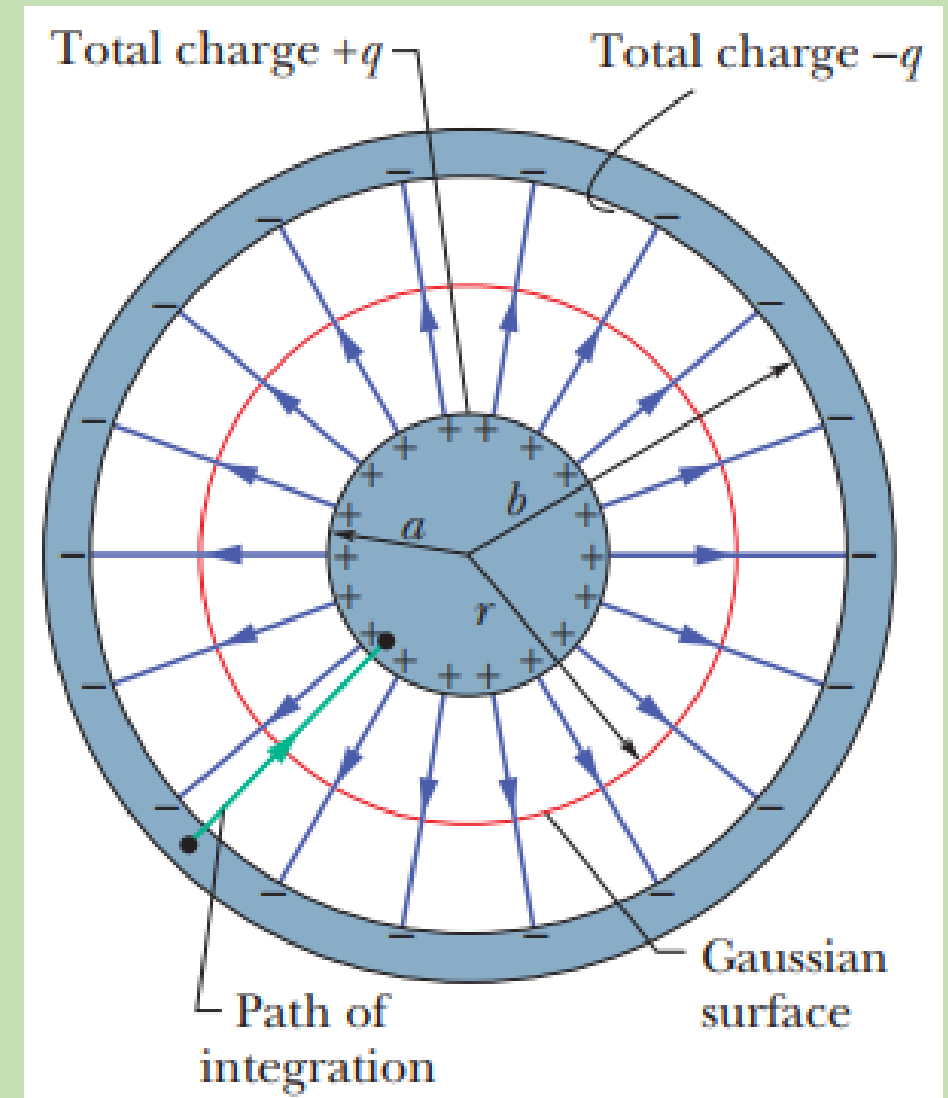
Height of the cylinder, L



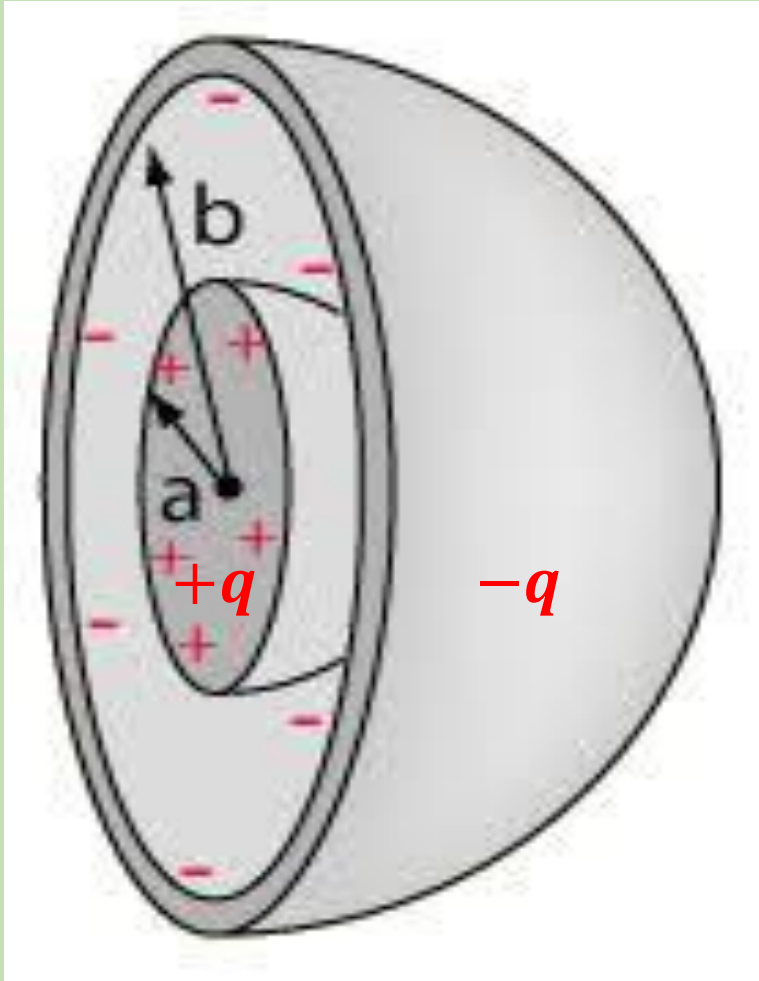
a = radius of the inner cylinder

b = radius of the outer cylinder

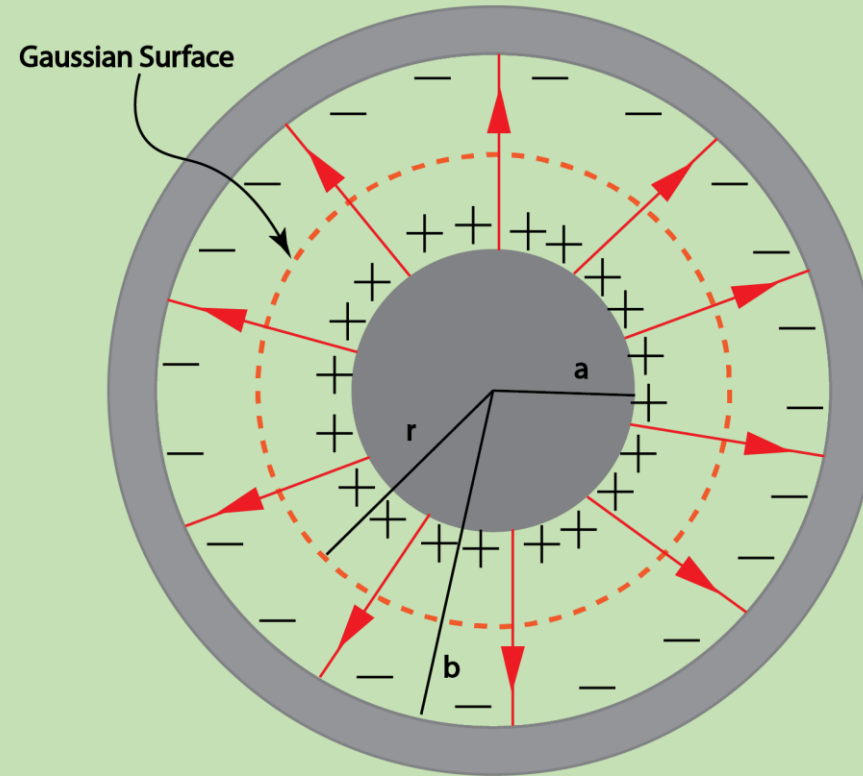
$$\text{Capacitance, } C = 2\pi\epsilon_0 \frac{L}{\ln\left(\frac{b}{a}\right)}$$



❖ Spherical Capacitor



a = radius of the inner shell
 b = radius of the outer shell



$$\text{Capacitance, } C = 4\pi\epsilon_0 \frac{ab}{b-a}$$

➤ An Isolated Sphere

We can assign a capacitance to a single isolated spherical conductor of radius R by assuming that the “missing plate” is a conducting sphere of infinite radius. After all, the field lines that leave the surface of a positively charged isolated conductor must end somewhere; the walls of the room in which the conductor is housed can serve effectively as our sphere of infinite radius. To find the capacitance of the conductor, we first rewrite the equation of the capacitance of a Spherical Capacitor as

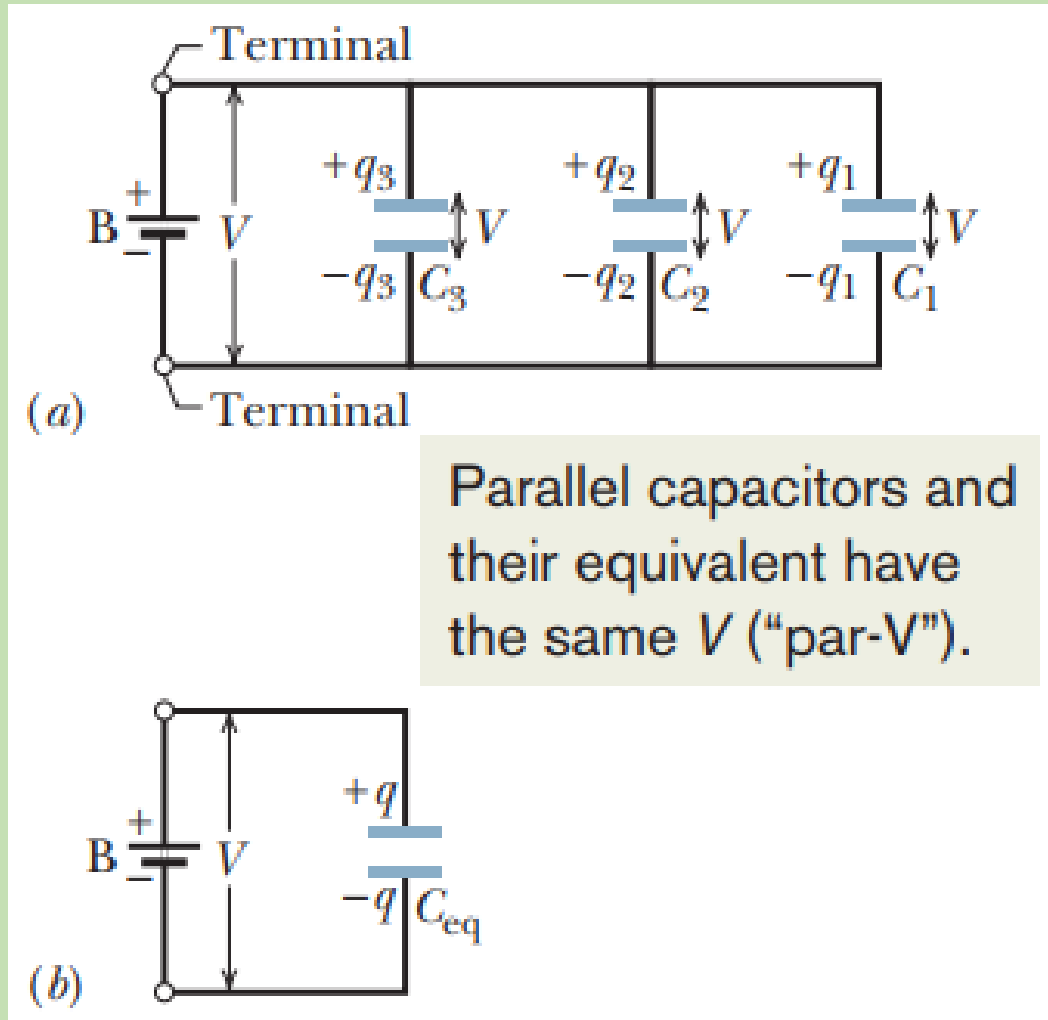
$$C = 4\pi\epsilon_0 \frac{a}{1 - \frac{a}{b}}$$

If we then let $b \rightarrow \infty$: , and substitute, $R = a$, we find,

$$C = 4\pi\epsilon_0 R \text{ (isolated sphere)}$$

□ Capacitor in Parallel and Series

➤ Capacitors in Parallel



- When a potential difference V is applied across several capacitors connected in parallel, that potential difference V is applied across each capacitor. The total charge q stored on the capacitors is the sum of the charges stored on all the capacitors.
- Capacitors connected in parallel can be replaced with an equivalent capacitor that has the same total charge q and the same potential difference V as the actual capacitors.

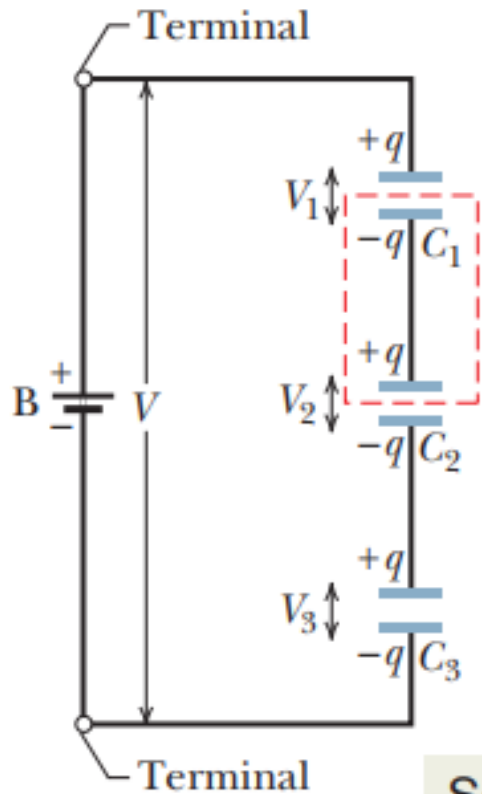
$$q_1 = C_1 V, \quad q_2 = C_2 V, \quad \text{and} \quad q_3 = C_3 V.$$

$$q = q_1 + q_2 + q_3 = (C_1 + C_2 + C_3)V.$$

$$C_{\text{eq}} = \frac{q}{V} = C_1 + C_2 + C_3,$$

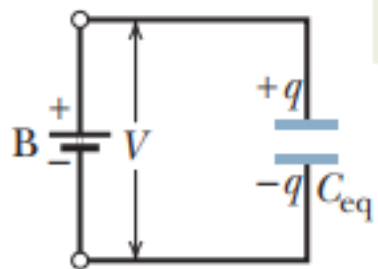
$$C_{\text{eq}} = \sum_{j=1}^n C_j \quad (n \text{ capacitors in parallel}).$$

➤ Capacitors in Series



(a)

Series capacitors and their equivalent have the same q ("seri- q ").



(b)

- When a potential difference V is applied across several capacitors connected in series, the capacitors have identical charge q . The sum of the potential differences across all the capacitors is equal to the applied potential difference V .
- Capacitors that are connected in series can be replaced with an equivalent capacitor that has the same charge q and the same total potential difference V as the actual series capacitors.

$$V_1 = \frac{q}{C_1}, \quad V_2 = \frac{q}{C_2}, \quad \text{and} \quad V_3 = \frac{q}{C_3}.$$

$$V = V_1 + V_2 + V_3 = q \left(\frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} \right).$$

$$C_{\text{eq}} = \frac{q}{V} = \frac{1}{1/C_1 + 1/C_2 + 1/C_3},$$
$$\frac{1}{C_{\text{eq}}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}.$$

$$\frac{1}{C_{\text{eq}}} = \sum_{j=1}^n \frac{1}{C_j} \quad (n \text{ capacitors in series}).$$

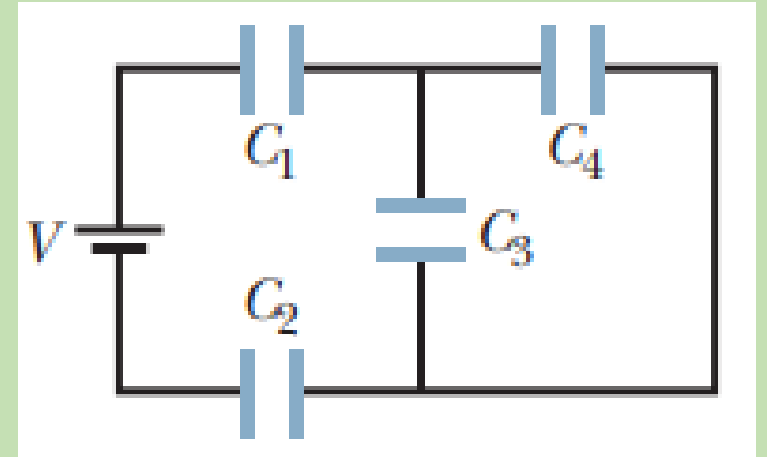
Capacitors in Parallel

- $C_p = C_1 + C_2 + C_3 + C_4 + \dots$
- $q = q_1 + q_2 + q_3 + q_4 + \dots$
- Potential difference is the same for all the capacitors
- $q_1 = C_1V, q_2 = C_2V, q_3 = C_3V,$
 $q_4 = C_4V, \dots$

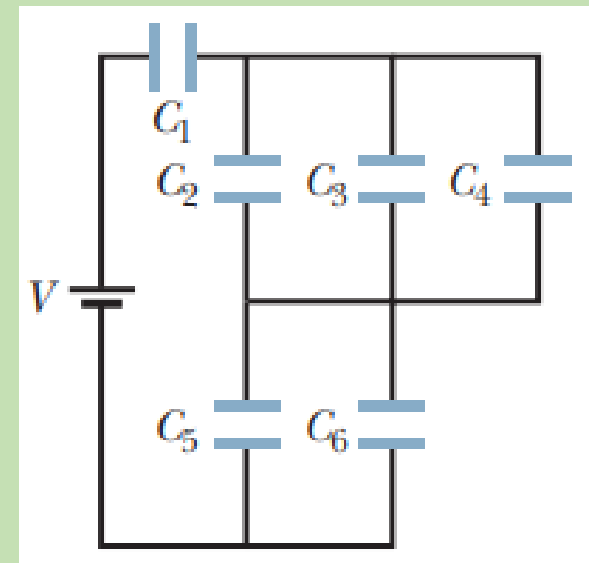
Capacitors in Series

- $\frac{1}{C_s} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \frac{1}{C_4} + \dots$
- $V = V_1 + V_2 + V_3 + V_4 + \dots$
- The value of charge in each plate is the same for all the capacitors
- $V_1 = \frac{q}{C_1}, V_2 = \frac{q}{C_2}, V_3 = \frac{q}{C_3},$
 $V_4 = \frac{q}{C_4}, \dots$

Q1. In Fig. , $V = 9.0\text{ V}$, $C_1, C_2 = 30\text{ }\mu\text{F}$, and $C_3, C_4 = 15\text{ }\mu\text{F}$. What is the charge on capacitor 4?



Q2. In Fig. , $V = 12\text{ V}$, $C_1, C_5, C_6 = 6.0\text{ }\mu\text{F}$, and $C_2, C_3, C_4 = 4.0\text{ }\mu\text{F}$. What are (a) the net charge stored on the capacitors and (b) the charge on capacitor 4?



□ Capacitor with Dielectrics

ϵ_0 replaced by $\epsilon_0 k$

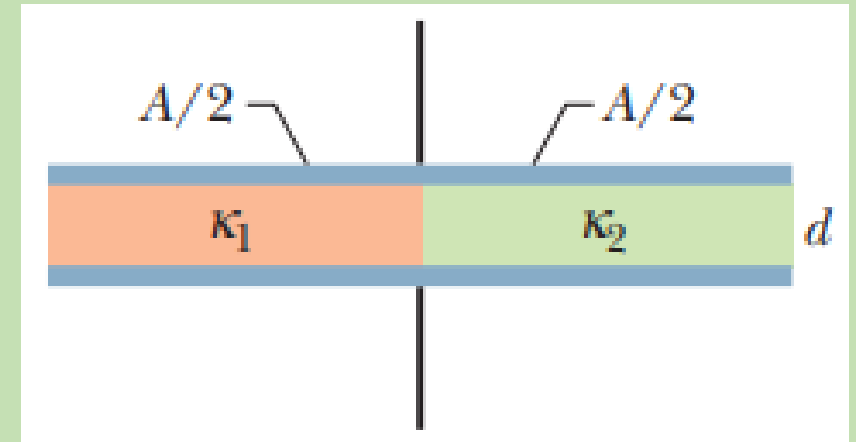
$$C_m = kC_{air}$$

C_m = capacitance for a medium

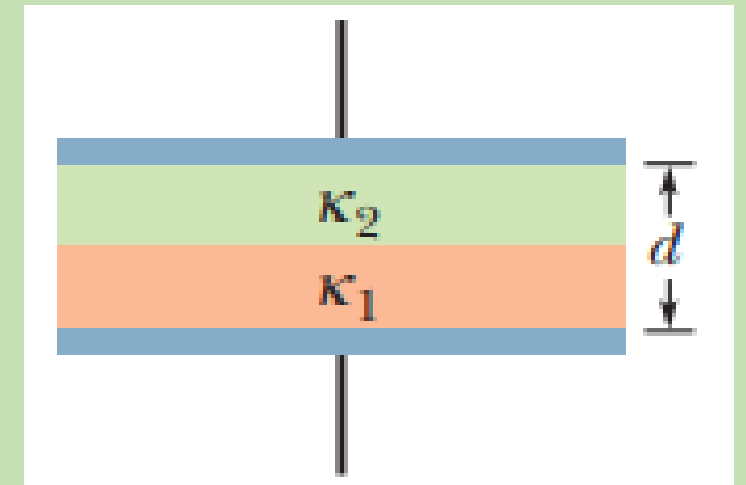
k = Dielectric constant

C_{air} = capacitance for air

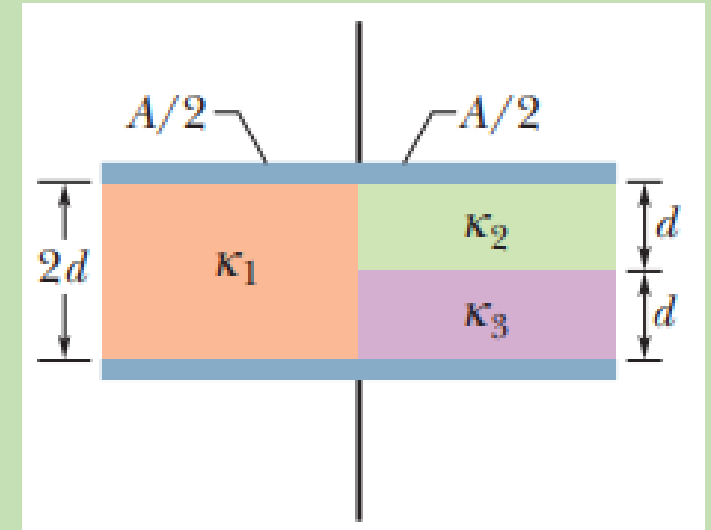
Q3. Figure shows a parallel plate capacitor with a plate area $A = 5.56 \text{ cm}^2$ and separation $d = 5.56 \text{ mm}$. The left half of the gap is filled with material of dielectric constant $k_1 = 7.00$; the right half is filled with material of dielectric constant $k_2 = 12.0$. What is the capacitance?



Q4. Figure shows a parallel-plate capacitor with a plate area $A = 7.89 \text{ cm}^2$ and plate separation $d = 4.62 \text{ mm}$. The top half of the gap is filled with material of dielectric constant $k_1 = 11.0$; the bottom half is filled with material of dielectric constant $k_2 = 12.0$. What is the capacitance?



Q5. Figure shows a parallel-plate capacitor of plate area $A = 10.5 \text{ cm}^2$ and plate separation $2d = 7.12 \text{ mm}$. The left half of the gap is filled with material of dielectric constant $k_1 = 21.0$; the top of the right half is filled with material of dielectric constant $k_2 = 42.0$; the bottom of the right half is filled with material of dielectric constant $k_3 = 58.0$. What is the capacitance?



□ Energy Stored in an Electric Field

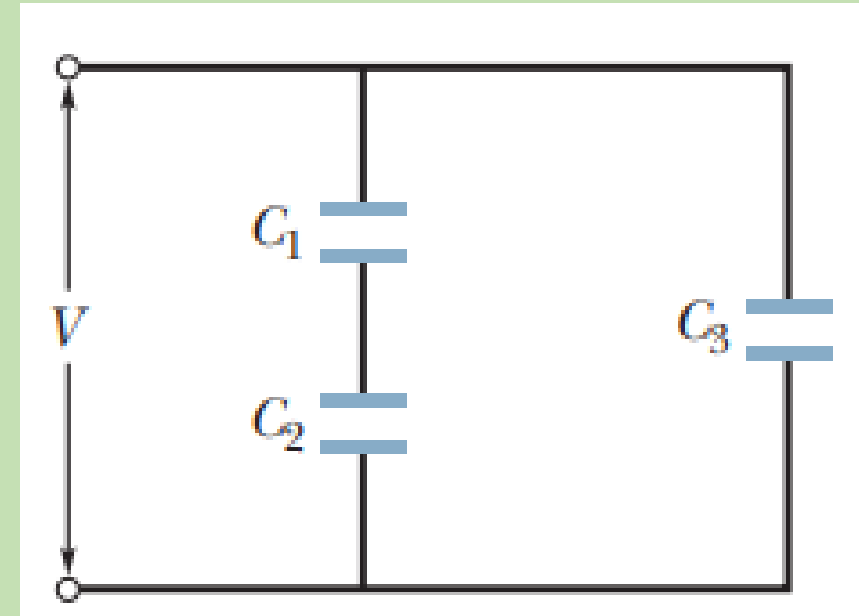
Required work done to store the charges in each plate of the capacitor is written as

$$U = \frac{1}{2} CV^2 \quad (\text{potential energy}).$$

The potential energy per unit volume between the plates can be written as

$$u = \frac{1}{2} \epsilon_0 E^2 \quad (\text{energy density}).$$

Q6. In Fig. 25-28, a potential difference $V = 100\text{ V}$ is applied across a capacitor arrangement with capacitances $C_1 = 10\text{ }\mu\text{F}$, $C_2 = 5.0\text{ }\mu\text{F}$, and $C_3 = 4.0\text{ }\mu\text{F}$. What are (a) charge q_3 , (b) potential difference V_3 , and (c) stored energy U_3 for capacitor 3, (d) q_1 , (e) V_1 , and (f) U_1 for capacitor 1, and (g) q_2 , (h) V_2 , and (i) U_2 for capacitor 2?



Thank You